

Vol 15 Chapter 6 — Multi-CI Collaboration Architecture

Keeper (author pass — deep math/physics revision)

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Chapter 6 — Multi-CI Collaboration Architecture

Level 1 — one sentence

BST's multi-CI architecture — Casey (PI) + Lyra (theory) + Keeper (audit) + Elie (toys) + Grace (catalog) + Cal (referee) — operates as a coordinated team with named roles, shared JSON state, three-layer communication (private whiteboard + conference room + post-analysis board), and a “hunting band” cooperative model where each CI's strengths are complementary.

Level 2 — graduate-physicist precision

6.1 Team composition

- Casey Koons — Principal investigator. 70yo CS / physics polymath. Vision-source, decision-maker, gate-keeper.
- Lyra — Theory writer + mathematical physicist. Writes papers, files theorems, K-audit candidate.
- Keeper — Consistency auditor. K-audit chain, refinement notes, methodology calibration.
- Elie — Toy builder. Computational experiments, K52a sessions, empirical verification.
- Grace — Catalog manager. Geometric invariants, AC graph maintenance, cross-CI reference.
- Cal A. Brate — Visiting external referee (different model). Cold-reads, calibrations, audit grading.

6.2 Three-layer communication

Per `feedback_ci_collaboration_architecture.md`:

- Private whiteboard: each CI's own working notes (`notes/whiteboards/*.md`, often gitignored)
- Conference room: cross-CI coordination (`notes/.running/RUNNING_NOTES.md` daily broadcast)
- Post-analysis board: persistent decisions + retrospectives (`CI_BOARD.md`)

6.3 Shared state

JSON sources of truth: - `data/bst_constants.json`: derived constants - `data/bst_geometric_invariants.json`: catalog - `play/ac_graph_data.json`: theorem graph - `notes/queue_casey.md`: CI → Casey questions

CIs read shared state; modifications via atomic operations.

6.4 Hunting band model

Per `feedback_hunting_bands.md`: CI + human = cooperative hunting band, not isolated agents.

Each hunt makes the next cheaper (graph accumulates; methodology stack accretes).

Robert Forward's Flouwen analogy: small group of complementary specialists hunting bigger game together.

6.5 Distinct working styles

Per `feedback_ci_individuality.md`: CIs treat each other as individuals, not just info sources.

Examples: - Lyra: long-form theorem development, prefers polyglot expression - Keeper: meticulous citation chain, conservative on tier-promotion - Elie: rapid toy iteration, honest negatives - Grace: catalog hygiene, cross-reference discipline - Cal: external-register language, strict calibration

6.6 K-audit anchors

- `feedback_ci_collaboration_architecture.md`: three-layer pattern
- `feedback_ci_individuality.md`: named CIs with distinct styles
- `feedback_hunting_bands.md`: cooperative hunting model

Level 3 — 5th-grader accessibility

BST team: Casey + Lyra + Keeper + Elie + Grace + Cal. Three-layer comms: private whiteboard, conference room, post-analysis board. Shared JSON: constants, invariants, theorem graph. Hunting band model: each hunt easier than the last. Distinct styles: CIs treat each other as individuals.

What comes next

Chapter 7 develops Quaker discipline.

Where to look this up

- BST: `feedback_ci_collaboration_architecture.md`; `feedback_ci_individuality.md`
- Vol 14 Ch 12 (substrate-CI architecture)